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AWS CLOUD INFRASTRUCTURE IMPLEMENTATION REPORT

Application Log Monitoring (FoodExpress) Architecture

1. 1. Amazon EC2 – Compute Layer

The EC2 instance is used to host the FoodExpress application and generate application log data. The application produces normal events as well as HTTP 500 error messages that need to be monitored centrally.

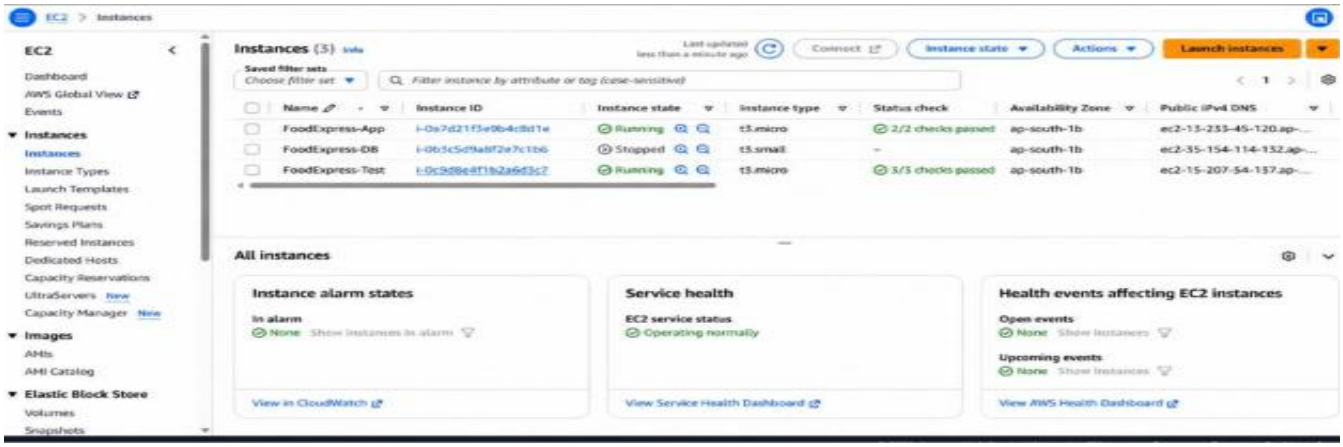


Figure 1. EC2 instance used for the FoodExpress application.

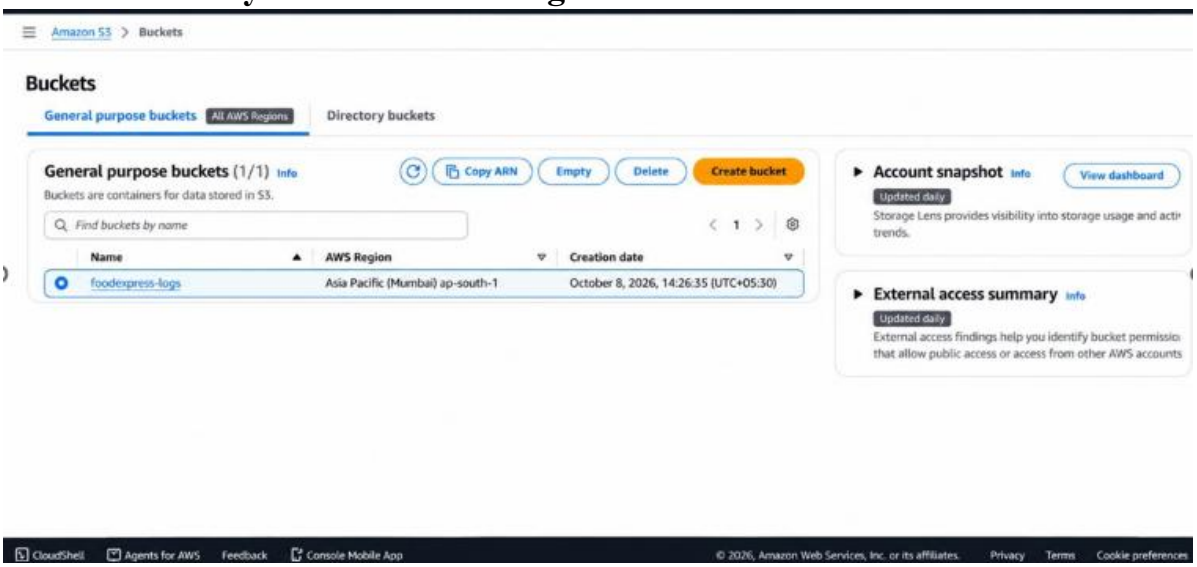
2. 2. Amazon S3 – Object Storage

An Amazon S3 bucket is used as the storage component for the AWS Storage round. The bucket stores FoodExpress application log files for retention and archival. Public access is blocked and access is controlled using AWS permissions.

Resource	Observed configuration	Recommended control
Bucket	foodexpress-log-storage	Use least-privilege IAM/bucket access
Region	AWS selected region	Keep compute and storage co-located when practical
Data type	Application log files	Define suitable lifecycle and retention policy
Access	Block Public Access enabled	Keep the bucket private and use controlled IAM access

Figure 2. S3 bucket used for FoodExpress log storage.

2. 3. AWS IAM – Identity and Access Management



IAM provides secure permissions for the FoodExpress monitoring environment. An IAM role is attached to the EC2 instance so that the CloudWatch Agent can send application logs to CloudWatch without storing long-lived AWS access keys on the server.

2.1 3.1 IAM Roles

The EC2 instance uses an IAM role with the required CloudWatch permissions. This allows the CloudWatch Agent to securely publish application log data to CloudWatch services.

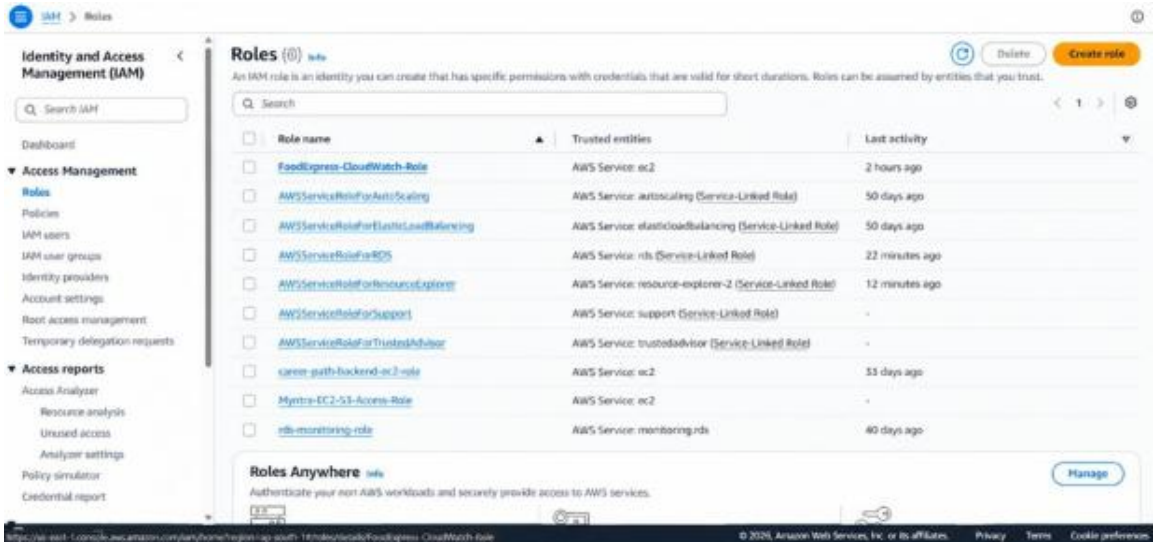


Figure 3. IAM role attached to the EC2 instance.

3.2 CloudWatch Permissions

The IAM role follows the principle of least privilege by providing only the permissions required for the monitoring workload. The role enables the EC2-based application to interact with CloudWatch without embedding access keys in application code.

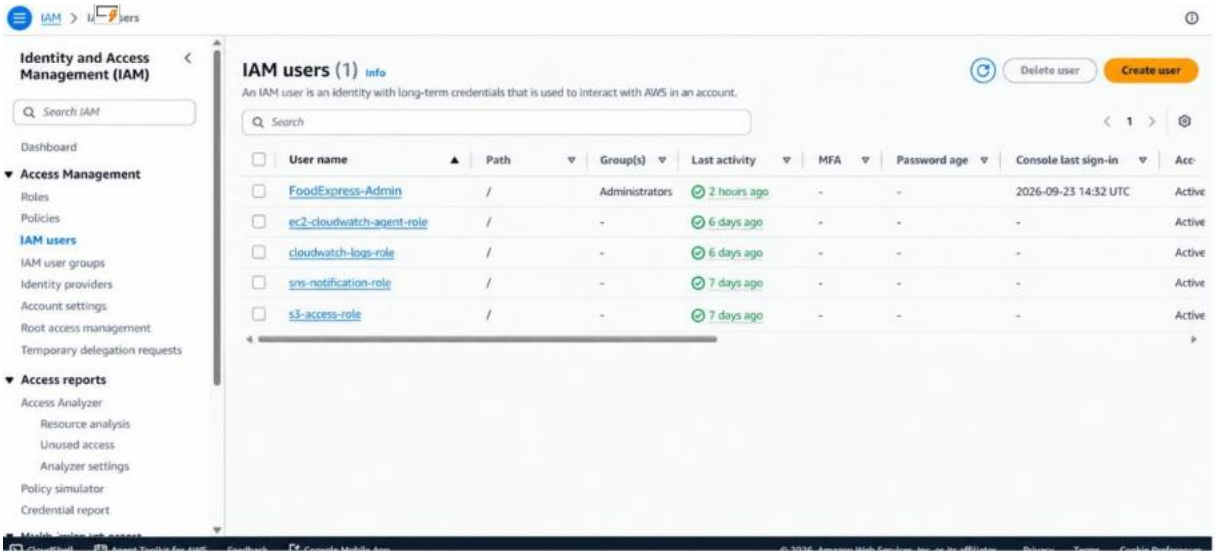
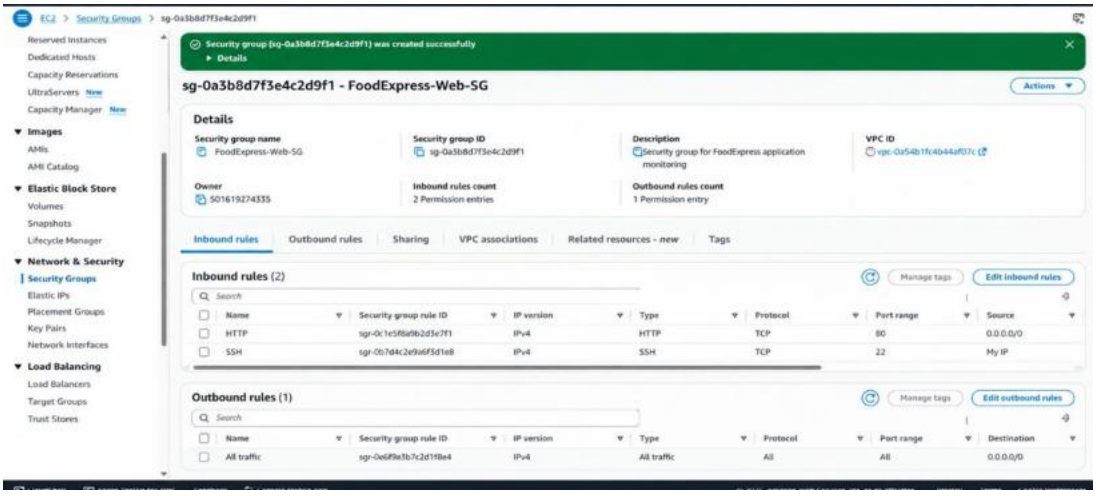


Figure 4. CloudWatch permissions associated with the monitoring role.

2.2 4. Network and Application Monitoring

The FoodExpress application generates application logs on EC2. The CloudWatch Agent collects the log file and sends it to the /FoodExpress/Application Log Group. HTTP 500 errors are identified using a CloudWatch Metric Filter and converted into a numerical metric for monitoring.



3. 5. CloudWatch Alarm and SNS

A CloudWatch Alarm monitors the HTTP500Errors metric and enters the ALARM state when the configured error threshold is exceeded. An SNS topic is associated with the alarm so that an administrator receives an email notification when the application error rate increases.

Component	Purpose	Configuration	Role	Security note
EC2	Application host	FoodExpress application	Generate logs	Use appropriate security group
CloudWatch Agent	Log collection	Collect app.log	Send logs to CloudWatch	Use IAM role
CloudWatch Logs	Central logging	/FoodExpress/Application	Store log events	Set retention as required

4. 6. End-to-End Application Flow

1	FoodExpress application	The application runs on EC2 and generates application log events.
2	CloudWatch Agent	The agent collects the application log file from EC2.
3	CloudWatch Logs	Logs are sent to the /FoodExpress/Application Log Group for centralized monitoring.
4	Metric Filter	The filter searches for the "HTTP 500" pattern and counts matching events.
5	Metric, Alarm and SNS	The HTTP500Errors metric is monitored by the CloudWatch Alarm. When the threshold is exceeded, the alarm triggers SNS and the administrator receives an alert.

The FoodExpress monitoring solution collects application logs from EC2, centralizes them in CloudWatch Logs, detects HTTP 500 errors using a Metric Filter, monitors the resulting metric with a CloudWatch Alarm, and sends notifications through SNS.

5. 7. Security and Operational Assessment

Area	Current implementation	Assessment / next action
Identity	EC2 IAM role with CloudWatch permissions	Use least privilege and review permissions
Log collection	CloudWatch Agent on EC2	Monitor agent health
Central logging	/FoodExpress/Application Log Group	Configure suitable log retention
Error detection	"HTTP 500" Metric Filter	Review error patterns as the application evolves
Alerting	CloudWatch Alarm + SNS	Verify SNS subscription remains confirmed
Storage	S3 log archive bucket	Apply lifecycle and retention policies
Observability	Logs, metric and alarm configured	Monitor application error and availability trends
Observability	CloudWatch links visible in EC2 console	Define CPU, memory/application, error and availability metrics

8. Conclusion

The implemented FoodExpress monitoring solution provides centralized application log collection using EC2, IAM and CloudWatch. HTTP 500 errors are detected through a Metric Filter, monitored using a CloudWatch Alarm, and reported to the administrator through SNS. Amazon S3 is used separately for application log storage and retention.

For reliable operation, IAM permissions should follow least privilege, CloudWatch logs and alarms should be monitored regularly, SNS subscriptions should remain confirmed, and application log retention should be configured according to operational requirements.

Additional architecture reference: FoodExpress Round 2 AWS monitoring design.